27. What is scatter diagram?

28. Define co-variance.

29. Define rank correlation.

30. If $\sum D^2 = 0$ what is your conclusion regarding Spearman's rank correlation coefficient?

31. Give an example for (i) positive correlation

(ii) negative correlation (iii) no correlation

32. What is the value of 'r' when two variables are uncorrelated?

33. When the correlation coefficient is +1, state your interpretation.

III. Give short answer to the following questions.

34. Write any three uses of correlation.

35. Define Karl Pearson's coefficient of correlation.

36. How do you interpret the coefficient of correlation which lies between 0 and +1?

37. Write down any 3 properties of correlation?

38. If rank correlation coefficient $r = 0.8$, $\sum D^2 = 3$ then find n?

39. Write any three merits of scatter diagram.

40. Given that cov$(x, y) = 18.6$, variance of $x = 20.2$, variance of $y = 23.7$. Find r.

41. Test the consistency of the following data with the symbols having their usual meaning. $N = 1000$, $(A) = 600$, $(B) = 500$, $(AB) = 50$.

IV. Give detailed answer to the following questions.

42. Explain different types of correlation.

43. Explain scatter diagram.

44. Calculate the Karl Pearson's coefficient of correlation for the following data and interpret.

x	9	8	7	6	5	4	3	2	1
y	15	16	14	13	11	12	10	8	9

45. Find the Karl Pearson's coefficient of correlation for the following data.

Wages	100	101	102	102	100	99	97	98	96	95
Cost of living	98	99	99	97	95	92	95	94	90	91

How are the wages and cost of living correlated?

46. Calculate the Karl Pearson's correlation coefficient between the marks (out of 10) in statistics and mathematics of 6 students.

Student	1	2	3	4	5	6
Statistics	7	4	6	9	3	8
Mathematics	8	5	4	8	3	6

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